Let us calculate the half life period for a zero order reaction.

Rate constant, $k = \frac{[A_0]-[A]}{t}$

at $t = t_{1/2}$; $[A] = [A_0]/2$

$$k = \frac{[A_0] - [A_0]/2}{t_{1/2}}$$

$$k = \frac{[A_0]}{2t_{1/2}}$$

$$t_{1/2} = \frac{[A_0]}{2k}$$

Hence, in contrast to the half life of a first order reaction, the half life of a zero order reaction is directly proportional to the initial concentration of the reactant.

More to know

Half life for an nth order reaction involving reactant A and $n \neq 1$

$$t_{1/2} = \frac{2^{n-1} - 1}{(n-1)k\,[A_0]^{n-1}}$$

Example 4

(i) A first order reaction takes 8 hours for 90% completion. Calculate the time required for 80% completion. (log 5 = 0.6989 ; log10 = 1)

Solution:

For a first order reaction,

$$k = \frac{2.303}{t}\log\left(\frac{[A_0]}{[A]}\right) \qquad \ldots(1)$$

Let $[A_0] = 100M$

When

$t = t_{90\%}$; $[A] = 10M$ (given that $t_{90\%} = 8$hours)

$t = t_{80\%}$; $[A] = 20M$

$$k = \frac{2.303}{t_{80\%}}\log\left(\frac{100}{20}\right)$$

$$t_{80\%} = \frac{2.303}{k}\log(5) \qquad \ldots(2)$$

Find the value of k using the given data

$$k = \frac{2.303}{t_{90\%}}\log\left(\frac{100}{10}\right)$$

$$k = \frac{2.303}{8\text{ hours}}\log 10$$

$$k = \frac{2.303}{8\text{ hours}}(1)$$

Substitute the value of k in equation (2)

$$t_{80\%} = \frac{2.303}{2.303/8\text{ hours}}\log(5)$$

$t_{80\%} = 8\text{hours} \times 0.6989$

$t_{80\%} = 5.59\text{hours}$

Example 5

(ii) The half life of a first order reaction $x \longrightarrow$ products is 6.932×10^4s at 500K . What percentage of x would be decomposed on heating at 500K for 100 min. ($e^{0.06} = 1.06$)

Solution:

Given $t_{1/2} = 0.6932 \times 10^4\,s$

To solve : when t=100 min,

$$\frac{[A_0]-[A]}{[A_0]} \times 100 = ?$$

We know that

For a first order reaction, $t_{1/2} = \frac{0.6932}{k}$

$$k = \frac{0.6932}{6.932 \times 10^4}$$

$k = 10^{-5}\,s^{-1}$

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